# Rates of Change & Concavity

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Average rate of change**

Find the average rate of change of $f(x) = x^2 + 2x$ on the interval $[1, 4]$.

AROC $= \frac{f(4) - f(1)}{4 - 1} = \frac{24 - 3}{3} = \frac{21}{3} = 7$.

**2** **[Foundational]** • **Target: Average rate of change**

A population grows according to $P(t) = 200(1.05)^t$ (years). Find the average rate of change of the population from $t = 0$ to $t = 10$, to the nearest whole number.

$P(0) = 200$. $P(10) = 200(1.05)^{10} \approx 200(1.6289) \approx 325.8$. AROC $= \frac{325.8 - 200}{10} \approx 12.6$ people/year.

**3** **[Foundational]** • **Target: Increasing/decreasing behavior from rate of change**

A function $f$ has average rates of change of $2$ on $[0, 1]$, $5$ on $[1, 2]$, and $9$ on $[2, 3]$. Determine whether $f$ appears to be concave up or concave down on $[0, 3]$, and explain.

Since the average rate of change is itself increasing ($2 \to 5 \to 9$), $f$ is speeding up as it increases — this is the signature of a concave-up function on $[0, 3]$.

**4** **[Foundational]** • **Target: Increasing/decreasing behavior from rate of change**

A different function $g$ has average rates of change of $8$ on $[0, 1]$, $5$ on $[1, 2]$, and $1$ on $[2, 3]$. Classify the concavity of $g$ on $[0, 3]$.

The average rate of change is decreasing ($8 \to 5 \to 1$), meaning $g$'s growth is slowing down — this indicates $g$ is concave down on $[0, 3]$.

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**5** [Foundational] • Target: Visualizing concavity and inflection

The graph shows $f(x) = x^3 - 3x$, which changes concavity at $x = 0$.

**a** Describe the concavity of the graph on $(-2, 0)$. On $(-2, 0)$, the curve bends downward (concave down) — the graph lies below its tangent lines there.

**b** Describe the concavity of the graph on $(0, 2)$. On $(0, 2)$, the curve bends upward (concave up) — the graph lies above its tangent lines there.

**c** State what kind of point $x = 0$ represents. $x = 0$ is an inflection point: the location where concavity changes from down to up.

(a) On $(-2, 0)$, the curve bends downward (concave down) — the graph lies below its tangent lines there.; (b) On $(0, 2)$, the curve bends upward (concave up) — the graph lies above its tangent lines there.; (c) $x = 0$ is an inflection point: the location where concavity changes from down to up.

**6** [Foundational] • Target: Rate of change of the rate of change

Using average rates of change over consecutive equal-width intervals as a proxy for how the function's growth rate itself is changing, explain what it means for a function to be "concave up" purely in terms of consecutive AROC values, without any calculus notation.

A function is concave up on an interval when its average rate of change is itself increasing as you move from left to right across consecutive equal-width sub-intervals — the function isn't just increasing, it's increasing at a faster and faster pace (or decreasing at a slower and slower pace).

**7** [Foundational] • Target: Applying rate-of-change reasoning to models

A company's revenue (in thousands) is modeled by $R(t) = t^3 - 6t^2 + 40t$ for $0 \le t \le 10$ years. Find the average rate of change on $[0, 2]$ and on $[2, 4]$, then determine whether revenue growth is accelerating or decelerating over $[0, 4]$.

$R(0) = 0$, $R(2) = 8 - 24 + 80 = 64$, $R(4) = 64 - 96 + 160 = 128$. AROC on $[0, 2]$: $\frac{64 - 0}{2} = 32$. AROC on $[2, 4]$: $\frac{128 - 64}{2} = 32$. Since both average rates are equal, revenue growth appears roughly linear (neither clearly accelerating nor decelerating) over this specific interval pair.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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**8** [Intermediate] • Target: Applying rate-of-change reasoning to models

For the same revenue function, find the AROC on $[4, 6]$ and compare it to the AROC on $[2, 4]$ found above, to determine the concavity trend as $t$ increases past $4$.

$R(6) = 216 - 216 + 240 = 240$. AROC on $[4, 6]$: $\frac{240 - 128}{2} = 56$. Since $56 > 32$ (the AROC on $[2, 4]$), the rate of change is increasing, indicating the revenue curve is concave up on $[4, 6]$ — growth is accelerating in this later interval.

**9** [Intermediate] • Target: Estimating rate of change from a table

A car's odometer reading (miles) is recorded every hour: $t = 0$: $120$; $t = 1$: $175$; $t = 2$: $238$; $t = 3$: $310$. Find the AROC over each hour and determine whether the car's speed is increasing, decreasing, or constant.

AROC on $[0, 1]$: $175 - 120 = 55$ mph. AROC on $[1, 2]$: $238 - 175 = 63$ mph. AROC on $[2, 3]$: $310 - 238 = 72$ mph. Since the AROC is increasing each hour ($55 \to 63 \to 72$), the car's speed is increasing — consistent with a concave-up distance function.

**10** [Intermediate] • Target: Estimating rate of change from a table

Using the same table, find the AROC over the full interval $[0, 3]$ and explain why it differs from the individual hourly rates.

AROC on $[0, 3]$: $\frac{310 - 120}{3} = \frac{190}{3} \approx 63.3$ mph. This is an average over the whole trip, blending the slower early hour with the faster later hour — it necessarily falls between the smallest (55) and largest (72) hourly rates, but doesn't equal any single hourly rate.

**11** [Intermediate] • Target: Visualizing rate of change on a graph

The graph shows $f(x) = \sqrt{x}$, whose average rate of change decreases as $x$ increases.

**a** Find the AROC of $f(x) = \sqrt{x}$ on $[0, 1]$ and on $[4, 9]$. AROC on $[0, 1]$: $\frac{\sqrt{1} - \sqrt{0}}{1} = 1$. AROC on $[4, 9]$: $\frac{\sqrt{9} - \sqrt{4}}{5} = \frac{3 - 2}{5} = 0.2$.

**b** Explain how the shrinking steepness in the graph relates to your two AROC values. The curve rises steeply near $x = 0$ and flattens out as $x$ increases, matching the drop from an AROC of $1$ down to $0.2$ — the function is growing, but at an ever-slower pace.

**c** State whether $f(x) = \sqrt{x}$ is concave up or concave down, based on this pattern. Since the AROC is decreasing as $x$ increases, $f(x) = \sqrt{x}$ is concave down on $(0, \infty)$.

(a) AROC on $[0, 1]$: $\frac{\sqrt{1} - \sqrt{0}}{1} = 1$. AROC on $[4, 9]$: $\frac{\sqrt{9} - \sqrt{4}}{5} = \frac{3 - 2}{5} = 0.2$.; (b) The curve rises steeply near $x = 0$ and flattens out as $x$ increases, matching the drop from an AROC of $1$ down to $0.2$ — the function is growing, but at an ever-slower pace.; (c) Since the AROC is decreasing as $x$ increases, $f(x) = \sqrt{x}$ is concave down on $(0, \infty)$.

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**12** **[Intermediate]** • **Target: Concavity and real-world interpretation**

A company reports that its costs are increasing, but at a decreasing rate. Sketch what this implies about the concavity of the cost function, and explain your reasoning without using calculus notation.

"Increasing but at a decreasing rate" means the function is going up (each AROC is positive) but each successive AROC is smaller than the one before — this is the definition of concave down. So the cost function's graph rises but bends downward, flattening out over time.

**13** **[Intermediate]** • **Target: Applying rate-of-change reasoning to models**

Find the average rate of change of the polar radius $r(\theta) = 3 + 2\sin(2\theta)$ over the interval $[0, \frac{\pi}{4}]$, and interpret whether the distance from the origin is increasing or decreasing on this interval.

Step 1: Evaluate $r(\theta)$ at $\theta = 0$ and $\theta = \pi/4$: $r(0) = 3 + 2\sin(0) = 3, \quad r(\frac{\pi}{4}) = 3 + 2\sin(\frac{\pi}{2}) = 3 + 2 = 5.$ Step 2: Compute the average rate of change: $\frac{r(\pi/4) - r(0)}{\pi/4 - 0} = \frac{5-3}{\pi/4} = \frac{8}{\pi} \approx \mathbf{2.546}$ units/rad. Because $r(\theta)$ is strictly increasing and positive on $[0, \pi/4]$, the curve moves further away from the pole.

**14** **[Intermediate]** • **Target: Estimating rate of change from a table**

Level 2 · MediumA table of values for a positive function $g(x)$ over equally spaced inputs $x = 0, 1, 2, 3$ produces outputs $g(0) = 4, g(1) = 12, g(2) = 36, g(3) = 108$. Which function family best models $g(x)$, and what is the constant growth factor per unit increase in $x$?

Calculate the consecutive ratios of consecutive outputs: $\frac{g(1)}{g(0)} = \frac{12}{4} = 3, \quad \frac{g(2)}{g(1)} = \frac{36}{12} = 3, \quad \frac{g(3)}{g(2)} = \frac{108}{36} = 3.$ Because the ratio of outputs is constant for equal-length input intervals, $g(x)$ is an exponential function with a constant growth factor of 3.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Visualizing rate of change on a graph** • **Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions**

Level 2 · MediumFind all real solutions to the trigonometric equation $2\cos^2\theta + 3\sin\theta - 3 = 0$ on the interval $[0, 2\pi)$.

Step 1 (Convert to Single Trig Ratio): Substitute $\cos^2\theta = 1 - \sin^2\theta$:

$2(1 - \sin^2\theta) + 3\sin\theta - 3 = 0 \Rightarrow 2 - 2\sin^2\theta + 3\sin\theta - 3 = 0. \quad -2\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0.$ Step 2 (Factor the Quadratic in $\sin\theta$): $(2\sin\theta - 1)(\sin\theta - 1) = 0.$ Step 3 (Solve the Linear Factors): $\sin\theta = 1 \Rightarrow \theta = \frac{\pi}{2}.2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}.$ In $[0, 2\pi)$.

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**16** **[Challenging] • Target: Concavity and real-world interpretation • Cross-Lesson: 4.03 Parametric Equations & Modeling**

Level 2 · MediumSolve the logarithmic equation $\log_2(x+3)+\log_2(x-1)=5$ analytically for all real solutions. Show the verification step that eliminates extraneous roots.

Step 1 (Apply Product Property): $\log_2((x+3)(x-1))=5$. Step 2 (Convert to Exponential Form): $(x+3)(x-1)=2^5=32$. Step 3 (Solve the Quadratic Equation): $x^2+2x-3=32\Rightarrow x^2+2x-35=0$. $(x+7)(x-5)=0\Rightarrow x=-7$ or $x=5$. Step 4 (Domain Verification): For $x=-7$: $x-1=-8\le 0$, and $\log_2(-8)$ is undefined in the real numbers. Hence $x=-7$ is an extraneous solution. For $x=5$: $5+3=8>0$ and $5-1=4>0$. Checking.

**17** **[Challenging] • Target: Average rate of change • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros**

A bivariate dataset $(x, y)$ is transformed into a semi-log plot $(x, \ln y)$, yielding the linear regression model $\ln(\hat{y})=1.5+0.4x$. Express $\hat{y}$ as an exponential function of $x$ in the form $\hat{y}=a\cdot b^x$, and find the predicted percentage growth rate per unit increase in $x$.

Step 1: Exponentiate both sides of the semi-log regression equation: $\hat{y}=e^{1.5+0.4x}=e^{1.5}\cdot(e^{0.4})^x$. Step 2: Evaluate parameters $a$ and $b$: $a=e^{1.5}\approx 4.4817$, $b=e^{0.4}\approx 1.4918\Rightarrow \mathbf{\hat{y}=e^{1.5}(e^{0.4})^x\approx 4.482(1.492)^x}$. Step 3: Percentage growth rate: Since $b-1=e^{0.4}-1\approx 0.4918$, $\hat{y}$ increases by approximately **49.18%** per unit increase in $x$.

**18** **[Challenging] • Target: Increasing/decreasing behavior from rate of change • Cross-Lesson: 1.02 Rational Functions & Asymptotes**

Level 1 · FoundationsA sample of a radioactive isotope has an initial mass of 250 mg and decays at a continuous rate of 4.8% per day according to $M(t)=250e^{-0.048t}$, where $t$ is measured in days. Determine the exact daily multiplicative decay factor $b$ in the form $M(t)=250b^t$, and calculate the daily percentage loss rounded to two decimal places.

Multiplicative Base: Rewrite $e^{-0.048t}=(e^{-0.048})^t$: $b=e^{-0.048}\approx 0.953134$. The exact base is $b=e^{-0.048}$ (or approximately 0.9531). Daily Percentage Loss: The percentage decay rate per day is $1-b$: $1-0.953134=0.046866\Rightarrow 4.69\%$. The daily percentage loss is 4.69% (distinguished from the continuous rate of 4.80%).

**19** **[Challenging] • Target: Visualizing concavity and inflection • Cross-Lesson: 2.01 Exponential Functions & Growth Models**

Solve the exponential equation $9^x-4\cdot 3^{x+1}+27=0$ for all real $x$.

Step 1: Rewrite in terms of $u=3^x$: Note that $9^x=(3^x)^2=u^2$ and $3^{x+1}=3\cdot 3^x=3u$: $u^2-12u+27=0$. Step 2: Factor the quadratic: $(u-3)(u-9)=0\Rightarrow u=3$ or $u=9$. Step 3: Back-substitute $3^x=u$: $3^x=3^1\Rightarrow x=1$; $3^x=3^2\Rightarrow x=2$. Solutions: $\mathbf{x=1,\,2}$.

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**20** [Challenging] • Target: Rate of change of the rate of change • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Level 1 · FoundationsLet $q(x) = -(x+2)^2(x-1)(x-4)^3$. What is the degree of $q(x)$, does the graph cross or touch the $x$-axis at $x = -2$, and what is $\lim_{x\to\infty} q(x)$?

Sum the exponents of the linear factors to find the degree: $2 + 1 + 3 = 6 \Rightarrow$ degree $6$. At $x = -2$, the multiplicity is 2 (even), so the graph touches the $x$-axis and turns around without crossing. The degree is even (6) and the leading coefficient is negative ($-1$), so $\lim_{x\to\infty} q(x) = -\infty$.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Average rate of change

Find the average rate of change of $f(x) = -7x^2$ on $[-4, -1]$.

**a.** $35$ ✓

**b.** $-21$

**c.** $33$

**d.** $37$

AROC $= \frac{f(-1) - f(-4)}{-1 - (-4)} = \frac{-7 - (-112)}{3} = 35$.

**22** [Foundational] • Target: Increasing/decreasing behavior from rate of change

A car's odometer reading (miles) is recorded every hour: $t = 0$: $131$; $t = 1$: $150$; $t = 2$: $178$; $t = 3$: $215$. Is the car's speed increasing, decreasing, or constant?

**a.** Decreasing

**b.** Constant

**c.** Cannot be determined from this data

**d.** **Increasing** ✓

AROC on $[0, 1]$: $19$ mph. AROC on $[1, 2]$: $28$ mph. AROC on $[2, 3]$: $37$ mph. The hourly rates are increasing, so the car's speed is Increasing.

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**23** [Foundational] • Target: Visualizing concavity and inflection

A function $h$ has average rates of change of $4$ on $[0, 1]$, $2$ on $[1, 2]$, and $1$ on $[2, 3]$. Classify the concavity of $h$ on $[0, 3]$.

**a.** Concave up

**b.** Cannot be determined from this data

**c.** Linear (neither concave up nor down)

**d.** **Concave down ✓**

The average rate of change is decreasing ($4 \to 2 \to 1$), so $h$ is concave down on $[0, 3]$.

**24** [Foundational] • Target: Rate of change of the rate of change

The rate of change of a quantity $N$ is proportional to its current value, modeled by the differential equation $\frac{dN}{dt} = kN$, where $k$ is a constant. If the initial population at time $t = 0$ is $N_0$, and at time $t = 5$ years, the population has grown to $2N_0$, what is the population at time $t = 15$ years, expressed in terms of $N_0$?

**a.** $6N_0$

**b.** $8N_0$ ✓

**c.** $7N_0$

**d.** $10N_0$

The differential equation $\frac{dN}{dt} = kN$ is a standard model for exponential growth. The solution to this differential equation is of the form $N(t) = N_0e^{kt}$, where $N_0$ is the initial population at $t = 0$. We are given that at $t = 5$ years, the population is $2N_0$. We can use this information to find the constant $k$. $N(5) = N_0e^{k \cdot 5} = 2N_0$ Divide both sides by $N_0$: $e^{5k} = 2$ Take the natural logarithm of both sides: $5k = \ln(2)$ So, $k = \frac{\ln(2)}{5}$. Now we need to find the popul.

**25** [Foundational] • Target: Applying rate-of-change reasoning to models

If $2 + 3i$ is a root of a real polynomial, so is:

**a.** $2 - 3i$ ✓

**b.** $-2 + 3i$

**c.** $-2 - 3i$

**d.** $3 + 2i$

Complex conjugate $2 - 3i$ must also be a root.

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**26** [Foundational] • Target: Estimating rate of change from a table

Let $P(x)$ be a monic quartic polynomial with real coefficients whose graph is tangent to the $x$-axis at $x = -3$ and $x = 7$. A horizontal line $y = k$ intersects the graph of $y = P(x)$ at four distinct points whose $x$-coordinates, when listed in increasing order, form an arithmetic progression. What is the value of $k$?

**a.** 400 ✓

**b.** 225

**c.** 256

**d.** 625

1. **Determine $P(x)$:** Because $P(x)$ is a monic quartic polynomial that is tangent to the $x$-axis at $x = -3$ and $x = 7$, it has roots of multiplicity at least 2 at both points. Since $\deg(P) = 4$, these multiplicities must be exactly 2: $P(x) = (x + 3)^2(x - 7)^2$ 2. **Exploit the axis of symmetry:** The line of symmetry lies halfway between $-3$ and 7: $h = \frac{-3+7}{2} = 2$ Let $u = x - 2$. Then $x + 3 = u + 5$ and $x - 7 = u - 5$. In terms of $u$, the polynomial becomes.

**27** [Foundational] • Target: Visualizing rate of change on a graph

A right-angled triangle ABC is shown with the right angle at vertex C. The length of side AC is 8 units and the length of side BC is 6 units. What is the value of $\sin(\angle BAC)$?

**a.** $\frac{3}{5}$ ✓

**b.** $\frac{4}{5}$

**c.** $\frac{3}{4}$

**d.** $\frac{4}{3}$

The triangle ABC is a right-angled triangle with the right angle at C. We are given AC = 8 and BC = 6. We want to find $\sin(\angle BAC)$. In a right-angled triangle, the sine of an angle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. First, we need to find the length of the hypotenuse AB using the Pythagorean theorem: $AB^2 = AC^2 + BC^2$. $AB^2 = 8^2 + 6^2$ $AB^2 = 64 + 36$ $AB^2 = 100$ $AB = \sqrt{100} = 10$ units. Now, we can find $\sin(\angle BAC)$.

**28** [Foundational] • Target: Concavity and real-world interpretation

A point P lies on the unit circle such that its terminal ray makes an angle of $\theta$ with the positive x-axis. If the x-coordinate of P is $\frac{1}{2}$, what is the y-coordinate of P? Assume $\theta$ is in the first quadrant.

**a.** $\frac{\sqrt{3}}{2}$ ✓

**b.** $\frac{\sqrt{2}}{2}$

**c.** $\frac{1}{2}$

**d.** $\sqrt{3}$

For a point P on the unit circle, its coordinates are given by $(\cos\theta, \sin\theta)$. We are given that the x-coordinate of P is $\frac{1}{2}$, so $\cos\theta = \frac{1}{2}$. Since P is on the unit circle, its coordinates satisfy the equation $x^2 + y^2 = 1$. Substituting the x-coordinate, we get $(\frac{1}{2})^2 + y^2 = 1$. This simplifies to $\frac{1}{4} + y^2 = 1$. Subtracting $\frac{1}{4}$ from both sides gives $y^2 = 1 - \frac{1}{4} = \frac{3}{4}$. Taking the square root of both sides, we get.

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** **[Intermediate]** • **Target: Average rate of change**

The point P is located on the unit circle at an angle of $\frac{\pi}{3}$ radians counterclockwise from the positive x-axis. What are the Cartesian coordinates $(x, y)$ of point P?

**a.** $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

**b.** $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ ✓

**c.** $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

**d.** $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

The unit circle is defined by the equation $x^2 + y^2 = 1$. For a point on the unit circle at an angle $\theta$ from the positive x-axis, the Cartesian coordinates are given by $x = \cos(\theta)$ and $y = \sin(\theta)$. In this case, the angle is $\theta = \frac{\pi}{3}$ radians. We know that $\cos(\frac{\pi}{3}) = \frac{1}{2}$ and $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$. Therefore, the Cartesian coordinates of point P are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

**30** **[Intermediate]** • **Target: Increasing/decreasing behavior from rate of change**

A point $P$ is located on the unit circle such that its angle with the positive x-axis is $\frac{2\pi}{3}$ radians. What are the coordinates of point $P$?

**a.** $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

**b.** $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ ✓

**c.** $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

**d.** $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

The coordinates of a point on the unit circle corresponding to an angle $\theta$ are given by $(\cos\theta, \sin\theta)$. In this case, $\theta = \frac{2\pi}{3}$ radians. We know that $\cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$ and $\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$. Therefore, the coordinates of point $P$ are $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

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**31** [Intermediate] • Target: Visualizing concavity and inflection

$\tan\theta = \frac{\text{opp}}{\text{adj}}$. In 3-4-5 triangle with opp=3,adj=4:

**a.** $\frac{3}{4}$ ✓

**b.** $\frac{4}{3}$

**c.** $\frac{3}{5}$

**d.** $\frac{5}{3}$

Applying Unit 3 · Trigonometric and Polar Functions principles, $\tan = \frac{3}{4}$ Therefore, the verified correct option is $\frac{3}{4}$.

**32** [Intermediate] • Target: Rate of change of the rate of change

In polar form, which equation represents a circle of radius 3 centred at the origin?

**a.** $r = 3$ ✓

**b.** $r = 3\theta$

**c.** $r = 3\cos\theta$

**d.** $\theta = 3$

r = 3 means every point is exactly 3 units from the origin — a circle of radius 3.

**33** [Intermediate] • Target: Applying rate-of-change reasoning to models

The graph of the quadratic function $f(x) = \frac{1}{2}x^2 - 2$ is shown in the figure below. How many distinct real solutions does the equation $(f \circ f \circ f)(x) = 0$ have?

**a.** 3

**b.** 4 ✓

**c.** 6

**d.** 8

We solve the iterated equation $f(f(f(x))) = 0$ layer by layer from the outside in. **Step 1: Outer Layer** Let $u = (f \circ f)(x)$. The equation becomes: $f(u) = 0 \Rightarrow \frac{1}{2}u^2 - 2 = 0 \Rightarrow u^2 = 4 \Rightarrow u = -2 \quad \text{or} \quad u = 2$ **Step 2: Middle Layer** Now set $v = f(x)$ and solve $f(v) = u$ for each value of $u$: - Case A: $f(v) = -2$ $\frac{1}{2}v^2 - 2 = -2 \Rightarrow \frac{1}{2}v^2 = 0 \Rightarrow v = 0$ Since $-2$ is the minimum value (vertex) of the parabola, this y.

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**34** [Intermediate] • Target: Estimating rate of change from a table

A particle's position in the Cartesian plane is described by the polar coordinates $(r, \theta)$, where $r$ is the distance from the origin and $\theta$ is the angle measured counterclockwise from the positive x-axis. The particle begins at $\theta_1 = \frac{\pi}{6}$ with a radial distance of $r_1 = 4$. It then moves to a new position such that its new angle is $\theta_2 = \frac{2\pi}{3}$ and the radial distance is $r_2 = 6$. If the displacement vector of the particle is $\mathbf{d} = \langle x, y \rangle$, what is the value of $x^2 + y^2$?

**a.** 13

**b.** 25

**c.** 49 ✓

**d.** 52

The initial position of the particle in Cartesian coordinates is $(x_1, y_1) = (r_1 \cos\theta_1, r_1 \sin\theta_1)$. Given $r_1 = 4$ and $\theta_1 = \frac{\pi}{6}$, we have: $x_1 = 4\cos(\frac{\pi}{6}) = 4 \cdot \frac{\sqrt{3}}{2} = 2\sqrt{3}$ and $y_1 = 4\sin(\frac{\pi}{6}) = 4 \cdot \frac{1}{2} = 2$. So, the initial position is $(2\sqrt{3}, 2)$. The final position of the particle in Cartesian coordinates is $(x_2, y_2) = (r_2 \cos\theta_2, r_2 \sin\theta_2)$. Given $r_2 = 6$ and $\theta_2 = \frac{2\pi}{3}$.

**35** [Intermediate] • Target: Visualizing rate of change on a graph

Consider a particle whose position in the complex plane at time $t$ (in seconds) is given by $z(t) = 3e^{i(\frac{\pi}{6}t + \frac{\pi}{3})}$. (a) Find the initial position of the particle at $t = 0$ in polar coordinates $(r, \theta)$, where $r \geq 0$ and $0 \leq \theta < 2\pi$. (b) Determine the angular velocity of the particle in radians per second. (c) If the particle starts at a point A and moves along a circular path, find the arc length traveled by the particle from $t = 0$ to $t = 4$ seconds.

**a.** (a) $(3, \frac{\pi}{3})$, (b) $\frac{\pi}{6}$ rad/s, (c) $2\pi$

**b.** (a) $(3, \frac{\pi}{6})$, (b) $\frac{\pi}{3}$ rad/s, (c) $\frac{4\pi}{3}$

**c.** **(a)** $(3, \frac{2\pi}{3})$**, (b)** $\frac{\pi}{6}$ **rad/s, (c)** $\frac{2\pi}{3}$ ✓

**d.** (a) $(3, \frac{\pi}{3})$, (b) $\frac{\pi}{3}$ rad/s, (c) $\frac{8\pi}{3}$

The position of the particle is given by $z(t) = 3e^{i(\frac{\pi}{6}t + \frac{\pi}{3})}$. This is in the polar form $z(t) = re^{i\theta(t)}$, where $r = 3$ and $\theta(t) = \frac{\pi}{6}t + \frac{\pi}{3}$. (a) The initial position at $t = 0$ is $z(0) = 3e^{i(\frac{\pi}{6}(0) + \frac{\pi}{3})} = 3e^{i\frac{\pi}{3}}$. In polar coordinates $(r, \theta)$, the radius is $r = 3$. The angle is $\theta(0) = \frac{\pi}{3}$. Since $0 \leq \frac{\pi}{3} < 2\pi$, the initial position is $(3, \frac{\pi}{3})$. (b) The angular velocit.

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**36** **[Intermediate]** • **Target: Concavity and real-world interpretation**

A vector $\mathbf{v}$ is defined by its component form $\langle 2a+1, 3-b\rangle$. This vector is then transformed by a translation represented by the vector $\mathbf{t} = \langle 5, -2\rangle$. The resulting transformed vector, $\mathbf{v}'$, is given as $\langle 12, 1\rangle$. Determine the values of the parameters $a$ and $b$.

**a.** **a = 3, b = -4** ✓

**b.** a = 4, b = 3

**c.** a = 3, b = 4

**d.** a = 4, b = -3

The original vector is $\mathbf{v} = \langle 2a+1, 3-b\rangle$. The translation vector is $\mathbf{t} = \langle 5, -2\rangle$. The transformed vector is $\mathbf{v}' = \langle 12, 5\rangle$. The transformation is given by $\mathbf{v}' = \mathbf{v} + \mathbf{t}$. Substituting the given vectors: $\langle 12, 5\rangle = \langle 2a+1, 3-b\rangle + \langle 5, -2\rangle$ $\langle 12, 5\rangle = \langle (2a+1)+5, (3-b)+(-2)\rangle$ $\langle 12, 5\rangle = \langle 2a+6, 1-b\rangle$ Equating the corresponding.

**37** **[Intermediate]** • **Target: Average rate of change**

A point P has polar coordinates $(r, \theta)$. When converted to Cartesian coordinates, P is $(3, 3\sqrt{3})$. If the angle $\theta$ is restricted to the interval $[0, 2\pi)$, find the value of $r$ and $\theta$ in radians.

**a.** $r = 6, \theta = \frac{\pi}{3}$ ✓

**b.** $r = 3\sqrt{2}, \theta = \frac{\pi}{4}$

**c.** $r = 6, \theta = \frac{\pi}{6}$

**d.** $r = 3, \theta = \frac{\pi}{3}$

The conversion formulas from polar coordinates $(r, \theta)$ to Cartesian coordinates $(x, y)$ are $x = r\cos\theta$ and $y = r\sin\theta$. We are given the Cartesian coordinates $x = 3$ and $y = 3\sqrt{3}$. We can find $r$ using the relationship $r^2 = x^2 + y^2$. $r^2 = 3^2 + (3\sqrt{3})^2$ $r^2 = 9 + (9 \times 3)$ $r^2 = 9 + 27$ $r^2 = 36$ Since $r$ represents a distance, it must be non-negative. Thus, $r = \sqrt{36} = 6$. Now we find $\theta$ using the relationships $\tan\theta = \frac{y}{x}$ (if $xeq0$).

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Increasing/decreasing behavior from rate of change • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Consider a linear transformation $T$ in the plane represented by the matrix $M = (a, b; c, d)$. A triangle with vertices at $A = (1, 2)$, $B = (3, 1)$, and $C = (2, 4)$ is transformed by $T$ to a new triangle $A'B'C'$. If the area of the original triangle $ABC$ is 3 square units and the area of the transformed triangle $A'B'C'$ is 18 square units, and given that $a = 2$ and $d = 3$, find the product of the possible values of the determinant of $M$.

**a.** 6

**b.** -6

**c.** 36

**d.** **-36** ✓

The area of the transformed triangle is related to the area of the original triangle by the absolute value of the determinant of the transformation matrix. That is, Area($A'B'C'$) = |det($M$)| * Area($ABC$). We are given Area($ABC$) = 3 and Area($A'B'C'$) = 18. Thus, $18 = |\det(M)| \cdot 3$, which implies $|\det(M)| = 6$. The determinant of $M$ is given by $\det(M) = ad - bc$. We are given $a = 2$ and $d = 3$, so $\det(M) = (2)(3) - bc = 6 - bc$. Since $|\det(M)| = 6$, we have two possibilities: $\det(M) = 6$ or.

**39** [Challenging] • Target: Visualizing concavity and inflection • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Consider a right-angled triangle ABC, where the right angle is at vertex B. Point D lies on the hypotenuse AC such that BD is perpendicular to AC. If AB = 10 units and BC = 24 units, find the length of AD.

**a.** $\frac{100}{26}$ ✓

**b.** $\frac{240}{26}$

**c.** $\frac{576}{26}$

**d.** $\frac{600}{26}$

Let the triangle be ABC with $\angle B = 90^\circ$. $BD \perp AC$, with D on AC. By the Pythagorean theorem in $\Delta ABC$, we have $AC^2 = AB^2 + BC^2$. Given $AB = 10$ and $BC = 24$, we get $AC^2 = 10^2 + 24^2 = 100 + 576 = 676$. So, $AC = \sqrt{676} = 26$ units. In a right-angled triangle, the altitude to the hypotenuse creates similar triangles. $\Delta ABC \sim \Delta ADB \sim \Delta BDC$. From the similarity $\Delta ABC \sim \Delta ADB$, we have the ratio of corresponding sides.

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**40** [Challenging] • Target: Rate of change of the rate of change • Cross-Lesson: 2.02 Logarithmic Functions & Properties

A circle is centered at the origin, and a point P lies on the circle. The angle formed by the positive x-axis and the line segment OP (where O is the origin) is $\theta$. The coordinates of point P are $(r\cos\theta, r\sin\theta)$. If point P has coordinates $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$ and lies on a circle of radius $R$, find the value of $R$.

**a.** $\frac{5}{2}$

**b.** 5 ✓

**c.** $\frac{5\sqrt{3}}{2}$

**d.** 10

The coordinates of a point P on a circle of radius R centered at the origin are given by $(R\cos\theta, R\sin\theta)$. We are given that the coordinates of P are $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$. Therefore, we can set up the following equations: $R\cos\theta = \frac{5}{2}$ $R\sin\theta = \frac{5\sqrt{3}}{2}$ To find R, we can square both equations and add them together, using the identity $\cos^2\theta + \sin^2\theta = 1$.

**41** [Challenging] • Target: Applying rate-of-change reasoning to models • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

A transformation is applied to a triangle ABC, mapping it to triangle A'B'C'. The transformation can be represented by a matrix multiplication of the form $M(x; y) = (3, 1; 1, k)(x; y)$, where $M$ is the transformation matrix and $k$ is an unknown parameter. If the original point A is $(2, 1)$ and its transformed image A' is $(7, 5)$, find the value of $k$.

**a.** 2

**b.** **3** ✓

**c.** 4

**d.** 5

The transformation matrix $M$ is given by $M = (3, 1; 1, k)$. We are given that point A $(2, 1)$ is transformed to point A' $(7, 5)$. This can be expressed as a matrix equation: $M(2; 1) = (7; 5)$. Substituting the matrix M, we get: $(3, 1; 1, k)(2; 1) = (7; 5)$. Performing the matrix multiplication for the first row: $(3 \cdot 2) + (1 \cdot 1) = 6 + 1 = 7$. This matches the first component of A', confirming the top row of the matrix is corre.

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**42** [Challenging] • Target: Estimating rate of change from a table • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

The terminal ray of an angle $\theta$ in standard position in the $xy$-plane passes through the point $(-5, 12)$. What is the value of $\sin\theta$?

**a.** $-\frac{5}{13}$

**b.** $\frac{12}{13}$ ✓

**c.** $-\frac{12}{5}$

**d.** $\frac{12}{5}$

For an angle $\theta$ in standard position whose terminal ray passes through a point $(x, y)$, the trigonometric functions are defined using the distance from the origin to the point, $r = \sqrt{x^2 + y^2}$. 1. Calculate $r$: $r = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ 2. Compute $\sin\theta$: $\sin\theta = \frac{y}{r} = \frac{12}{13}$ Common misconceptions: - $-\frac{5}{13}$ is $\cos\theta = \frac{x}{r}$. - $-\frac{12}{5}$ is $\tan\theta = \frac{y}{x}$. - $\frac{12}{5}$ incorrectly uses.

**43** [Challenging] • Target: Visualizing rate of change on a graph • Cross-Lesson: 4.01 Inverse & Composite Functions

A vector $\mathbf{v}$ is defined by its initial point $P = (-5, 3)$ and its terminal point $Q = (4, -2)$. What is the component form of vector $\mathbf{v}$?

**a.** $\langle -9, 5\rangle$

**b.** $\langle 9, -5\rangle$ ✓

**c.** $\langle -1, 1\rangle$

**d.** $\langle 1, -1\rangle$

The component form of a vector $\mathbf{v}$ directed from an initial point $P = (x_1, y_1)$ to a terminal point $Q = (x_2, y_2)$ is given by $\mathbf{v} = \langle x_2 - x_1, y_2 - y_1\rangle$. In this case, the initial point is $P = (-5, 3)$ and the terminal point is $Q = (4, -2)$. Therefore, the component form of $\mathbf{v}$ is: $\mathbf{v} = \langle 4 - (-5), -2 - 3\rangle = \langle 4 + 5, -5\rangle = \langle 9, -5\rangle$.

**44** [Challenging] • Target: Concavity and real-world interpretation • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. Find the component form of $\mathbf{v}$.

**a.** $\langle 3, 4\rangle$ ✓

**b.** $\langle 7, 10\rangle$

**c.** $\langle -3, -4\rangle$

**d.** $\langle 4, 3\rangle$

The component form of a vector $\mathbf{v}$ with initial point $(x_1, y_1)$ and terminal point $(x_2, y_2)$ is given by $\mathbf{v} = \langle x_2 - x_1, y_2 - y_1\rangle$. In this case, $(x_1, y_1) = (2, 3)$ and $(x_2, y_2) = (5, 7)$. Therefore, the component form of $\mathbf{v}$ is $\langle 5 - 2, 7 - 3\rangle = \langle 3, 4\rangle$.

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**45** [Challenging] • Target: Average rate of change • Cross-Lesson: 4.03 Parametric Equations & Modeling

A transformation maps point $A = (1, 2)$ to $A' = (-4, 2)$. If this transformation is a translation, what is the component form of the translation vector $\mathbf{v}$?

**a.** $\langle -5, 0 \rangle$ ✓

**b.** $\langle 5, 0 \rangle$

**c.** $\langle -3, 0 \rangle$

**d.** $\langle 3, 0 \rangle$

Let the initial point be $A = (x_1, y_1)$ and the translated point be $A' = (x_2, y_2)$. The translation vector $\mathbf{v}$ is given by $\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, $(x_1, y_1) = (1, 2)$ and $(x_2, y_2) = (-4, 2)$. Therefore, $\mathbf{v} = \langle -4 - 1, 2 - 2 \rangle = \langle -5, 0 \rangle$.

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